

Laplace Transform of a Periodic Function

If $f(t)$ is periodic with period T , then:

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt$$

Proof: $\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} dt$

$$= \int_0^T e^{-st} f(t) dt + \int_T^{\infty} e^{-st} f(t) dt$$

let $u = t - T$ in the 2nd integral

then $du = dt$ and $\begin{cases} t = \infty \rightarrow u = \infty \\ t = T \rightarrow u = 0 \end{cases}$

$$\Rightarrow \mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt + \int_0^{\infty} e^{-s(u+T)} f(u+T) du$$

since f has period T , then

$$f(u+T) = f(u)$$

$$\Rightarrow \mathcal{L}\{f(t)\} = \int_0^T e^{-st} f(t) dt + \int_0^{\infty} e^{-s(u+T)} f(u) du$$

$$\Rightarrow F(s) = \int_0^T e^{-st} f(t) dt + e^{-sT} \underbrace{\int_0^{\infty} e^{-su} f(u) du}_{F(s)}$$

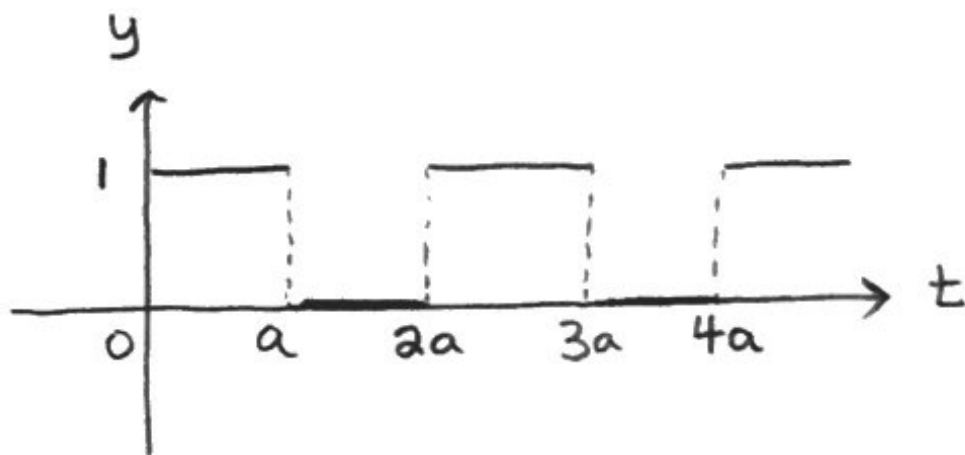
$$\Rightarrow F(s) = \int_0^T e^{-st} f(t) dt + e^{-sT} F(s)$$

$$\Rightarrow F(s) = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt$$

$$\Rightarrow \boxed{\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt}$$

ex: ① Find the Laplace Transform

of the square wave :



$$T = 2a$$

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-2as}} \int_0^{2a} e^{-st} f(t) dt$$

$$= \frac{1}{1 - e^{-2as}} \int_0^a e^{-st} dt$$

Note: The function is 0
between $t=a$ and $t=2a$

$$\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-2as}} \left[\frac{e^{-st}}{-s} \right] \Bigg|_{t=0}^{t=a}$$

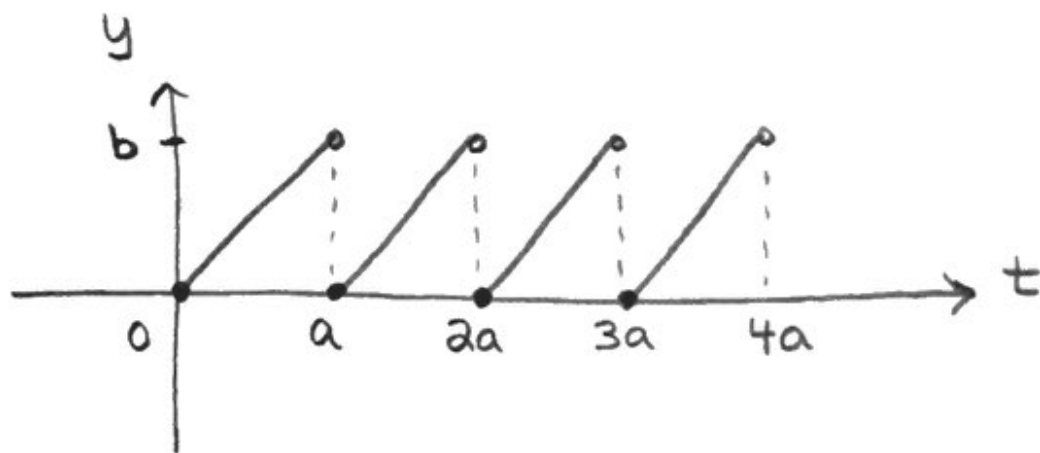
$$= \frac{1}{1-e^{-2as}} \left[\frac{e^{-as}}{-s} - \frac{e^0}{(-s)} \right]$$

$$= \frac{1-e^{-as}}{s(1-e^{-2as})}$$

$$= \frac{(1-e^{-as})}{s(1-e^{-as})(1+e^{-as})}$$

$$= \boxed{\frac{1}{s(1+e^{-as})}}$$

ex: (2) Find the Laplace Transform
of the sawtooth function:



$T = a \leftarrow$ Period

$$f(t) = \frac{bt}{a} \quad \text{over } 0 \leq t < a$$

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-as}} \int_0^a e^{-st} \left(\frac{bt}{a} \right) dt$$

$\swarrow$
we can use integration by
Parts for this one.

$$\mathcal{L}\{f(t)\} = \frac{b}{a(1-e^{-as})} \int_0^a t e^{-st} dt$$

D	I
t	$\oplus e^{-st}$
1	$\ominus -\frac{1}{s} e^{-st}$
0	$\oplus \frac{1}{s^2} e^{-st}$

$$\mathcal{L}\{f(t)\} = \frac{b}{a(1-e^{-as})} \left[-\frac{t}{s} e^{-st} - \frac{1}{s^2} e^{-st} \right] \bigg|_{t=0}^{t=a}$$

$$\mathcal{L}\{f(t)\} = \frac{b}{a(1-e^{-as})} \left[-\frac{ae^{-as}}{s} - \frac{e^{-as}}{s^2} + \frac{1}{s^2} \right]$$

$$= \frac{b}{a(1-e^{-as})} \left[-\frac{ae^{-as}}{s} + \frac{1-e^{-as}}{s^2} \right]$$

$$\mathcal{L}\{f(t)\} = \frac{b}{as} \left[\frac{-ae^{-as}}{1-e^{-as}} + \frac{(1-e^{-as})}{s(1-e^{-as})} \right]$$

$$= \frac{b}{as} \left[\frac{1}{s} - \frac{ae^{-as}}{1-e^{-as}} \right]$$

$$= \frac{b}{as} \left[\frac{1}{s} - \frac{a}{e^{as}(1-e^{-as})} \right]$$

$$= \boxed{\frac{b}{as} \left[\frac{1}{s} - \frac{a}{e^{as}-1} \right]}$$
